

CHAPTER

ELECTRIC FIELD

ticle is at certain S.
At this instant

Q. 1. Define and explain electric field in detail?

ELECTRIC FIELD

The space or region around a charge in which attraction Or repulsion with other charges takes place is called electric field.

The charge produces electric field which is denoted by dots. The density of dots is proportional to the strength of electric field. The presence of field is tested by test charge. charge is generally $+q$.

Consider a charge q which produces an electric field. The electric field of charge q interacts with test and gives rise to electric force. The electric force per unit test charge is called electric field

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OF ELECTRIC FIELD LINES

The electric field is a vector quantity and electric field lines give its direction. The electric field lines come out from positive charge and enter into negative charge, The two electric field lines never cross each other.

INIT

Q' 2' What are electric-field line of force? How are these lines drawn? Write their properties?

MAGNETISM

Now $-q$ charge at the centre of sphere. The test $+q_0$ is placed at 'any' direction of resultant electric field lines are given by the tangents of the sphere. Hence, electric field is same as shown in fig. field lines

PROPERTIES OF ELECTRIC FIELD

O The el•NtrlC posililR Charges x gative

The number of lines of force passing perpendicularly through an unit area gives the electric field at any point.

proportional to the magnitude of the electric field.

The electric field lines associated with two identical positive charges are curved and seem to repel each other. The middle region shows zero field location called neutral zone.

The behavior of electric field lines of two identical negative charges is also exactly the same. The field lines are curved which enter into negative charge. The middle region shows zero field location called neutral zone.

The electric field pattern of two opposite charges of same magnitude is - 'dipole' fig. The field lines start from positive charge and end on a negative charge. It is called electric field and denoted by E .

ELECTRIC FIELD DUE TO A POINT CHARGE

Consider a point charge q at a point P . The electric field E at a point P is defined as the force F experienced by a unit positive charge q_0 placed at P . Its direction is the direction of the force on a positive charge.

The magnitude of the electric field E at a point P due to a point charge q is given by:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

where r is the distance from the charge q to the point P .

Q.4 Calculate electric field due to continuous charge distribution.

Electric charge is quantized. It is a collection of large number of charges. Continuous charge distribution has three types: Linear charge distribution, Surface charge distribution, and Volume charge distribution.

- Linear charge distribution
- Surface charge distribution
- Volume charge distribution

ELECTRIC FIELD DUE TO LINEAR CHARGE DISTRIBUTION

Consider a linear charge distribution. Take a small length element dl of the distribution. The charge on this element is dq .

The electric field lines are curved. The electric lines rather than straight lines.

OR

Calculate electric field due to point charge and due to "point charge."

ELECTRIC FIELD DUE TO A POINT CHARGE

Consider a charge q placed at point O called point charge. Place a charge q_0 at point P having distance r from charge q . Electric force on q_0 is

dq length element has charge dq .

The charge per unit length is called linear charge density and denoted by λ .

The net electric field at P due to whole distribution is

OR this

The charge area is

field p due to charge dq is

net electric field at P due to whole surface charge distribution

ELECTRIC FIELD DUE TO VOLUME CHARGE DISTRIBUTION

Consider a volume charge distribution. Take a small volume element dV or this

Unit volume element has charge dq .

Unit Volume element has

dq

Charge unit is
and denoted by q

dV

electric at P due to is

at P

What is an electric dipole? calculate electric field
A pair of q distance $2a$ negative of same magnitude
them is the of charge are.
and in atoms solid and liquid materials

ELECTRIC dipole.

The product of q and separation between them is.

moment Vector from Charge to Charge
ELECTRIC DUE TO DIPOLE

Consider a charge
having q

them. is called
dipole moment
given as $p = q \cdot 2a$

Take a point P having r which
bisects $2a$ as shown in

The magnitude of electric field at
P due to $+q$ charge has $\frac{1}{r^2}$
distance r is

The direction of E is from $+q$ charge towards P. The magnitude of E at P
due to $-q$ charge having distance r

Now to calculate net electric field at P due to dipole.
Resolve the electric field

into

$$E \cos \theta + E \cos \theta$$

of is

(fig.

(From

the term in by theorem

Neglect the higher

get

and E into components.

Rectangular components of E are

$$E \cos \theta + E \cos \theta = 2E \cos \theta$$

The direction of E is from P towards $-q$ charge.

Comparing (1) and eq(2) we can write as

This is the electric field P due to electric dipole p is
fig. •

Rectangular components

$$E \sin \theta$$

of E are

into

and bisecting point O of $2a$ $+q$ $-q$

Q.6 Calculate the electric field due to ring of charge?**ELECTRIC FIELD DUE TO RING OF CHARGE**

Consider a positively charged ring having radius R on which positive charge is distributed uniformly. This is called linear charge distribution. Take a small length element ds of ring having charge dq .

The linear charge density λ is defined as

$$\lambda = \frac{dq}{ds}$$

Now consider two length elements ds at opposite ends of a diameter of ring. We have to calculate electric field at P which is on perpendicular axis at distance z from plane of ring. From fig-

$$r^2 = z^2 + R^2$$

The magnitude of electric field at P due to charge element L is,

$$dE_L = \frac{k dq}{r^2} \quad \text{----- (1)}$$

The magnitude of electric field at P due to charge element M is

$$dE_M = \frac{k dq}{r^2} \quad \text{----- (2)}$$

Comparing eq (1) and eq (2) we get

$$dE_L = dE_M$$

Now to calculate electric field dE at P due to both length elements, resolve electric field dE_L and dE_M into components.

Rectangular components of dE_L are

$$dE_{Lx} = dE_L \cos \theta, \quad dE_{Ly} = dE_L \sin \theta$$

Rectangular components of dE_M are

$$dE_{Mx} = dE_M \cos \theta, \quad dE_{My} = dE_M \sin \theta$$

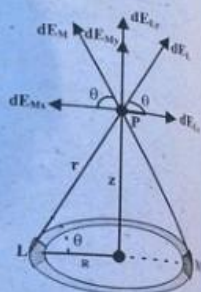
Resultant x -component of electric field is

$$dE_x = (+dE_{Lx}) + (+dE_{Mx})$$

$$dE_x = dE_L \cos \theta + dE_M \cos \theta$$

$$dE_x = dE_L \cos \theta + dE_L \cos \theta$$

$$dE_x = 0$$



Resultant y -component of electric field is

$$dE_y = (+dE_{Ly}) + (+dE_{My})$$

$$dE_y = dE_L \sin \theta + dE_M \sin \theta$$

$$dE_y = dE_L \sin \theta + dE_L \sin \theta$$

$$dE_y = 2 dE_L \sin \theta$$

The magnitude of electric field dE is

$$dE^2 = dE_x^2 + dE_y^2$$

$$dE^2 = (0)^2 + (2 dE_L \sin \theta)^2$$

$$dE = 2 dE_L \sin \theta$$

The net electric field at P due to whole ring of charge is

$$E = \int 2 dE_L \sin \theta$$

$$E = \int 2 \frac{k dq}{r^2} \left(\frac{z}{r} \right) \quad \left[\text{From fig. } \sin \theta = \left(\frac{z}{r} \right) \right]$$

$$E = \int 2 \frac{k dq}{r^2} \left(\frac{z}{r} \right) = \int \frac{2 z k dq}{r^3}$$

$$E = \frac{2 z k \lambda}{(z^2 + R^2)^{3/2}} \int ds$$

($\int ds = \pi R$) Length of circumference of half ring because length elements are taken at both sides of diameter.

$$E = \frac{2 z k \lambda}{(z^2 + R^2)^{3/2}} (\pi R)$$

As $\lambda = dq/ds$

$$\text{so } q = \lambda \int ds = \lambda (2\pi R)$$

$$E = \frac{z k q}{(z^2 + R^2)^{3/2}}$$

This is electric field at point P due to ring of charge. When point P lies at large distance $z \gg R$. The term R^2 can be neglected as compared to z^2

$$E = \frac{z k q}{(z^2 + 0)^{3/2}}$$

$$E = \frac{k q}{z^2}$$

It means ring behaves as a point charge which is concentrated at center of ring when $z \gg R$.

Q.7 Calculate electric field due to disk of charge.**ELECTRIC FIELD DUE TO DISK**

Consider a positively charged circular disk of radius R having surface charge density σ . Divide the disk into small rings. Now consider such a small ring having radius r and with $d\sigma$.

Take a pair of small area element da at opposite ends of diameter denoted by L and M. The area of the element having width da and length $S = \omega da$ is given as

$$da = \omega da \cdot d\omega$$

as

The area element da has charge dq . The charge per unit area is called surface charge density

$$\sigma = \frac{dq}{dA}$$

$$dq = \sigma dA$$

$$dq = \sigma(\omega da \cdot d\omega)$$

We have to calculate electric field at point P having distance z from the plane of the disk along its axis as shown in fig.

The electric field at P due to area element L having distance r is

$$dE_L = \frac{k dq}{r^2} \quad \text{--- (1)}$$

Similarly electric field at P due to area element M having distance r is

$$dE_M = \frac{k dq}{r^2} \quad \text{--- (2)}$$

Comparing eq (1) and eq (2), we get

$$dE_L = dE_M$$

Now to calculate electric field dE at P due to both area elements, resolve the electric field dE_L and dE_M into components.

Rectangular components of dE_L are

$$dE_{Lx} = dE_L \cos\theta \quad \text{and} \quad dE_{Ly} = dE_L \sin\theta$$

Rectangular components of dE_M are

$$dE_{Mx} = dE_M \cos\theta \quad \text{and} \quad dE_{My} = dE_M \sin\theta$$

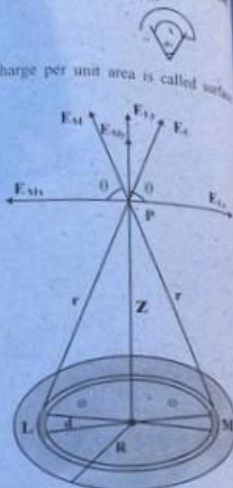
Resultant x-component of electric field dE_x is

$$dE_x = (+dE_{Lx}) + (+dE_{Mx})$$

$$dE_x = dE_L \cos\theta + dE_M \cos\theta$$

$$dE_x = dE_L \cos\theta + dE_L \cos\theta$$

$$dE_x = 0$$



Resultant y-component of electric field dE_y is

$$dE_y = (+dE_{Ly}) + (+dE_{My})$$

$$dE_y = dE_L \sin\theta + dE_M \sin\theta$$

$$dE_y = dE_L \sin\theta + dE_L \sin\theta$$

$$dE_y = 2 dE_L \sin\theta$$

The magnitude of electric field dE is

$$dE^2 = dE_x^2 + dE_y^2 = (0)^2 + (2 dE_L \sin\theta)^2$$

$$dE = 2 dE_L \sin\theta$$

$$dE = 2 \frac{k dq}{r^2} \left(\frac{z}{r} \right) \quad \left(\text{From fig. } \sin\theta = \frac{z}{r} \right)$$

$$dE = 2 \frac{k z dq}{r^3}$$

Use $dq = \sigma(\omega da \cdot d\omega)$ and from fig. $r^2 = z^2 + \omega^2$

$$dE = 2 \frac{k z \sigma \omega da \cdot d\omega}{(z^2 + \omega^2)^{3/2}}$$

To calculate the resultant field due to whole disk, integrate the above eq.

$$E = \int dE$$

$$E = \int \frac{k z \sigma \omega da \cdot d\omega}{(z^2 + \omega^2)^{3/2}}$$

$$E = k \sigma z \int_0^R \frac{2 \omega da \cdot d\omega}{(z^2 + \omega^2)^{3/2}}$$

The limit for radius of disk is from 0 to R. This R must be rotated through 2π for complete contribution of charge of disk. The limit for da is taken from 0 to π i.e. for half circle because contribution of other half circle is already added as area element is taken at both ends of diameter.

$$E = k \sigma z (\pi) \int_0^R \frac{(z^2 + \omega^2)^{-3/2} (2 \omega da \cdot d\omega)}{1}$$

$$E = k \sigma z (\pi) \left[\frac{(z^2 + \omega^2)^{-1/2}}{-\frac{3}{2} + 1} \right]_0^R$$

$$E = k \sigma z (\pi) \left(\frac{2}{1} \right) \left[\frac{1}{\sqrt{z^2 + \omega^2}} \right]_0^R$$

AND

ELECTRIC FIELD DUE TO INFINITE LINE OF CHARGE

Consider infinite line of positive charge having linear charge density λ at length element dl having charge dQ . The linear charge density is λ . Resultant E at point P is

Now consider pair of small length elements dl_1 and dl_2 each of length dl on the line. The electric field dE_1 and dE_2 at point P is given by Coulomb's law. The resultant electric field dE at point P is the vector sum of dE_1 and dE_2 . The electric field dE at point P is radially outward from the line. The electric field dE at point P is given by

$$dE = \frac{k \lambda dl}{r^2}$$

$$dE \cos \theta = \frac{k \lambda dl}{r^2} \cos \theta$$

$$dE \sin \theta = \frac{k \lambda dl}{r^2} \sin \theta$$

The magnitude of electric field dE is

The total electric field due to whole line is

integrate
When $z = 0$

$\frac{k \lambda}{z}$
Sec 20 do

$$(\sin\theta - \sin\theta')$$

Sino

$$F d \sin\theta$$

$$qEd \sin\theta$$

$$p E \sin\theta$$

The electric torque in is

Q.9 Define an electric dipole moment & derive an for acting on it when placed in a uniform electric field. work done external electric med in turning the dipole from initial angle to angle & potential energy.

ELECTRIC DIPOLE

A pair of charge negative Charge Of Same having distance. between is called dipole,

ELECTRIC DIPOLE MOMENT

The product of magnitude of charge and separation between them is called moment and as $p = q \cdot d$. It is a vector quantity. The direction of dipole vector is from negative charge to positive charge.

MOTION OF ELECTRIC DIPOLE IN ELECTRIC FIELD

Consider a q and $-q$ charge having distance d between them is placed in electric field E . The electric dipole moment vector P makes angle θ with electric

force on edge

The direction of electric torque is given by right hand rule which is directed

to right hand rule. Now vector P when released rotates clockwise

again tends to align with Hence electric torque and

The field produces a torque on dipole which to take to its energy configuration. Therefore, its rotation from the low energy state against field

The amount of work done by electric torque in electric field in turning the dipole through small angle $d\theta$ is

$$dw = \tau \cdot d\theta$$

The net work done in turning the dipole from initial position to final is

electric generated to couple

of electric acting dipole

as shown in fig.

AND

SHORT QUESTIONS

Charge a long silk thread. We wish to insert in the plane that or the hanging charge. To do put test charge q at that point and measure W/q . than, or

$$W =$$

arc

arc

is given by right hand rule,

the direction of the field?

test

"as

We Do have this Often really make a

charge but is of field while exploring electric

If is the direction of field lines is as "electric from and enter into Charge.

If is negative the field lines is given as "electric field from charge and enter positive

Electric field lines never cross each other

electric lines of force by Faraday. that charge is at the

Of a Circle, When test charge is placed at any circumference of circle, the

test charge moves along radial direction repulsive Therefore electric field lines Of circle The two tangents of a circle never cross each other so electric field lines also

$$-p \cos\theta - \cos\theta' \cdot w$$

in charge is equal to

that electric field are unidirectional. For curved field lines, the tangent to a point on any line gives the direction of the electric field line at that point. No two lines. This is because E has only one direction at any point.

E could have direction.

point charge is moving in an electric field at right angles to the electric lines of force. Does any force act on it.

h: The electrostatic force acts on charge when charge is moving in an electric field at right angle to the electric lines of force because electric is given

$$F = qE \cos\theta$$

What is the origin of electrostatic cling? a phenomenon that some times affects clothes as they are removed from dryer?

"v: cling is the which are due sharing of electrons. The charges are produced due to friction.

of dipole is 90° . The energy at any position is

$$-pE \cos\theta$$

$$-pE \cos\theta - \cos\theta' \cdot w$$

minimum when electric field E and dipole

$$-pE \cos\theta - \cos\theta' \cdot w$$

are anti parallel i.e, $\theta = 180^\circ$

POTENTIAL ENERGY

is

The potential energy will
vector $\vec{P}$ are parallel i.e $O =$

The

